

torch.diagonal_scatter#

https://pytorch.org/docs/stable/generated/torch.diagonal_scatter.html

Description:

Embeds the values of the src tensor into input along the diagonal elements of input, with respect to dim1 and dim2. This function returns a tensor with fresh storage; it does not return a view. The argument offset controls which diagonal to consider: If offset = 0, it is the main diagonal. If offset > 0, it is above the main diagonal.

Function Signature

```
torch.diagonal_scatter(input, src, offset=0, dim1=0, dim2=1)  
→ Tensor#
```

Parameters

Code Examples

```
>>> a = torch.zeros(3, 3)
>>> a
tensor([[0., 0., 0.],
        [0., 0., 0.],
        [0., 0., 0.]])

>>> torch.diagonal_scatter(a, torch.ones(3), 0)
tensor([[1., 0., 0.],
        [0., 1., 0.],
        [0., 0., 1.]])

>>> torch.diagonal_scatter(a, torch.ones(2), 1)
tensor([[0., 1., 0.],
        [0., 0., 1.],
        [0., 0., 0.]])
```

Notes & Warnings

NOTE

Note src must be of the proper size in order to be embedded into input. Specifically, it should have the same shape as `torch.diagonal(input, offset, dim1, dim2)`

Detailed Documentation

Documentation

Embeds the values of the src tensor into input along the diagonal elements of input, with respect to dim1 and dim2.

This function returns a tensor with fresh storage; it does not return a view.

The argument offset controls which diagonal to consider:

If offset = 0, it is the main diagonal.

If offset > 0, it is above the main diagonal.

If offset < 0, it is below the main diagonal.

input (Tensor) – the input tensor. Must be at least 2-dimensional.

src (Tensor) – the tensor to embed into input.

offset (int, optional) – which diagonal to consider. Default: 0 (main diagonal).

dim1 (int, optional) – first dimension with respect to which to take diagonal. Default: 0.

dim2 (int, optional) – second dimension with respect to which to take diagonal. Default: 1.

src must be of the proper size in order to be embedded into input. Specifically, it should have the same shape as `torch.diagonal(input, offset, dim1, dim2)`

